

Configuration-level split aliasing in object-detection benchmarks: what the reported split is selected on, and how much of the selection survives an independent split

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Provenance of this file. The English submission form of the P2 manuscript, a faithful

translation of the Chinese governing draft 评测有效性稿_正文_v0.1.md

(md5 fe880ca0e3af77983382065db75255f3, the state it was translated from, 2026-09-18; that draft is a live

document, re-hashed whenever it changes — see `work/verify_text_hashes_20260917.py`); every number is unchanged and backed by the

local artifacts listed in the Supplementary Material. Where the drafts disagree, the Chinese governs.

Citation numbers refer to the References section of this manuscript, whose numbering is not re-sorted

when entries are added later. Every entry was verified against Crossref or DataCite DOI records

or against the cited project's own page before it was added.

The four citations unresolved in the first draft — [A] the CVPR 2026 geospatial leakage

audit, [B] AI-TOD, [C] UAVDT, [D] D-Fire — are **now verified** as [1] [2] [3] [4]; two of

the four titles and one venue had differed in an unverified citation.

Abstract

Object-detection papers report a number from a split they call held out. We audit that premise, with three results.

1. The convention is broken in our own corpora. 6 of 14 YOLO configurations we produced give the reported number no independent held-out split: the reported checkpoint was selected on that split.

Training and evaluation do not overlap at the image level (intersection zero, SHWD excepted): not

contamination but **checkpoint selection**.

2. The direction varies by corpus. The selection premium is **positive in all four cells** (§5.1, an escalated four-cell base),

the baseline arm gaining more, while the paired contrast changes sign by corpus; a declared

13-cell family leaves **3** cells after Benjamini–Hochberg; in 19 public benchmarks the reported number

can come from a non-independent split (§6.1 — **one rater's judgement, not a rate**), yet each counting unit yields a defensible count. **Changing the metric keeps the mean direction, but one cell loses

significance and sign unanimity (‘**smoke2sf**’ +0.063 pp, **p = 0.54**, 6+/5–**).

3. Partial realization on a disjoint split. Separating the selection set from the reported set, that premium survives at an **arm-cell median 17.6 %**, span **–28.5 % to +73.7 % (102.2 pp)**,

2/5 negative; run-weighted **23.3 %**, at the claim's level [**–17.3 %**, **53.2 %**] (P = 0.118; **zero inside**),

run-level [8.4 %, 37.4 %] (P = 0.002); leave-one-out **9.77 %–34.16 %**.

Data-order replication lowers the mean **9 %–29 %** (§8); no prospective power analysis was registered, so the

nine n = 3 cells stay descriptive.

Keywords: evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit;

reproducibility

1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not

chosen on that same split. Binding `val:` and `test:` to one directory breaks it silently — the pipeline evaluates

there, keeps the best-scoring checkpoint and reports its score: a **maximum**, not a sample. Surveys of the detection literature [5] trace how that number became the

field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**,

where we count how many of the YOLO configurations we produced alias the two keys; **19 public

benchmarks, **audited row by row for which split the reported number comes from; and 20** published

third-party training logs**, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **measure the premium's fate** under a protocol

that separates the selection set from the reported set (best and last checkpoints and both splits exist, so the full 2×2 grid is computable): how much of the val-selected premium is realized on an

independent split — the measurement this paper contributes and §3.1 formalises. The protocol itself is the

companion paper's and is **cited, not restated**. And we **pre-register** a generalization hypothesis, then report that it misses its criterion — a registration **also reported in the companion**.

1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is

defensible and they disagree by a factor of six: 2/19 (the release artifact binds both keys to one path),

10/19 (no independent held-out test), 13/19 (the official protocol does not let you evaluate an independent

test locally), **12–14 of the 19 (a single rater's judgement, re-coded by models, not blind humans; the reported number comes from a non-independent split)**. These are defensible answers, not the four counting units (§6.1), whose unit-level counts are 10/19, 13/19, 4/19 and 12/19. Declaring the unit is

therefore a methodological premise, not a caveat (§3, §6).

1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the training images and the held-out evaluation images is **zero** (SHWD excepted, where train and val are one directory). We do not claim a single bias direction: our paired analysis finds **two** corpora

where val understates the effect and two where it overstates it^{**}, nine cells not significant (§5.1). We

do not claim the convention is framework-general (§6.5), nor that the benchmarks we audit are wrong: they

report what their configuration computes.

Positioning (one sentence). We claim neither to be first to audit split problems in detection benchmarks

nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level**

aliasing, a direction that varies by corpus, and the requirement that a count declare its unit.^{**} Every number points to a local artifact; every claim narrowed after external checking is

recorded as narrowed (§5.4).

Scope. All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**, the only

family among the six frameworks we checked (five pinned repositories plus the Ultralytics release) selecting the checkpoint by the validation metric by default

(§6.5) — stated at the claim, not only in a limitations section.

2. Contributions and related work

2.1 The closest work, and where this paper sits

Table S1 → Supplementary §S8 — verified against both 2026 studies' **full texts**, with the PDFs and extraction script archived.

None of these works is wrong — **they audit data (image identity, or the splitting unit), we audit configurations** (which path each key points to). Configuration-level aliasing shifts the

comparison between arms, whose direction we measure as **not single-valued** — a weaker statement

than [6], which finds rankings reversing across budgets on **all** of its benchmarks. Here ****2** of 13 cells are underestimated on val, 2 overestimated and 9 are not significant**, so no more than that is

claimed. They instantiate the taxonomy of [7], of which duplication is one mismatch [8]; binding val:

and test: to one directory creates the same mismatch **without duplicating an image**, so de-duplication cannot find it.

Non-collision, checked by literal search: three exact queries return **0** hits and the channel is **metadata-only** — which establishes that these strings are unused, **not that the idea is new** (Supplementary S12).

Why this is a selection question, not only a leakage question. The bias we measure is the one the

model-selection literature has described for decades: ****choosing a model by its score on a validation set**

makes the reported score optimistic [9], **as does the** choice of validation criterion** — accuracy-based

early stopping being the worst rule tested [10] — and the classical remedy is to *nest* the selection inside the evaluation rather than to de-duplicate the data. In detection the selection step is implicit:

the training script writes best.pt by fitness and the paper reports that checkpoint, often from a single

run [11]. A reader who trusts the split names therefore sees a held-out evaluation where the pipeline

performed a **maximum** — the bias the benchmark-erosion literature [12] describes, reached here with no

data reused or mis-split. We take [9] as the general statement and report the detection-specific

instantiation: **the selection premium (best – last), and how much of it transfers to a disjoint split**

(§5, §8).

The sharpest contrast is the two closest works: *Technologies* 2026 [13], which audits a detection

pipeline's **splitting unit** (frame versus sequence), and *Drones* 2026 [14], whose six configurations differ between val

and test by **between -0.41 and $+0.49$ mAP@0.5**, "with no systematic direction". We measure a ****different**

quantity: **on three-way corpora we** measure the paired difference's sign. **We therefore do not****

write "understated" unqualified, and state the difference from [14]: [14] **names the mechanism** — it writes that "the validation split is what early stopping and checkpoint selection saw" — but reports a raw val-minus-test

gap, mixing two splits' difficulty, **not identifiable**, whereas our **paired difference** cancels

difficulty and still shows a residual shift. The supporting reference [1] is not competing: cited only

as evidence that the genre has a venue.

That separation is the paper's own contribution: the literature above measures how much of the choice survives as a level [15] rather than predicting that in these

curves the premium is **dominated by the peak's position, not by run noise** (§8.6) — so its remedies are

not interchangeable with ours, and ****a val-side difference** stays uninterpretable as a generalisation

difference even after selection is removed**.

2.2 Contributions

What this paper reports first (the boundary to the companion is in §2.3): the configuration-level audit

of the val/test convention and its selection premium (§4–5); and the **measured** realization of that

premium on a disjoint split with the reporting conventions it depends on (§8.6–8.7, §11, §3.1).

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium

(best – last) is **positive in all four cells, the baseline arm gaining more** (four paired 95 % CIs excluding zero, §5.1 Tier 1) — **consistent with the selection-bias literature** [9], not new in

kind. **All four cells meet the stated-n criterion ($n = 10\text{--}11$)** — the fourth, dota15, entered at $n = 3$

and was escalated to $n = 10$ under a criterion fixed in advance (Supplementary S6, §5.1) — so the convention

systematically shifts **arm-to-arm comparisons**; we then **quantify the direction it takes**, which

varies by corpus: three-way paired differencing gives 2 cells where val understates, 2 where it overstates and 9 not significant — the **no-systematic-direction** [14] reports, measured here on detection corpora (the crossing cells are now **all at n = 10**).

We do not claim a general understatement (§5). It appears in third-party logs too (a searched 20-log sample; §5.5) — a **convenience sample found by repository search**, cited as corroboration that the component exists elsewhere, **not as independent evidence** and not as a field-level rate.

Note: image-level leakage magnitude is quantified by the two 2026 studies;*

configuration-level aliasing magnitude has not been measured before.*

2. **We measure how far the selection premium transfers.** The measurement is the contribution; the

protocol that makes it readable is the companion's and is **cited, not restated** here (§8.1). On the five arm-cells measured here the val-selected premium is realized on the disjoint split at

23.3 % with all five arm-cells at n = 10, and it is **not a constant**: the rate falls with the

validation noise of the criterion used to select (§8.6–§8.7). The identity tying it to the

winner's-curse share is one we **adopt** [16], not one we claim; the closest concurrent work [15, 17] reaches the same **direction** with a different quantity, and neither reports a **ratio of difference premiums** — $(T_{\text{best}} - T_{\text{final}})/(V_{\text{max}} - V_{\text{final}})$, on disjoint splits, in detection, on our own arm-cells, with measured val noise as moderator. A same-estimand ratio does exist elsewhere: in genomics the relative bias is written $1 - pG\text{-TS}/pG\text{-ES}$ and measured at 10–60 % (**Würschum & Kraft**, *Heredity* 112:463, 2014). The replicate axis is **data order** (§8.4) — the ****seed-budget discipline**

reproduced inside this setting** rather than a new one.

3. **We disclose our own defects** rather than removing them: our analysis' first version had non-reproducible items — scripts depending on a rented pod, a citation to a file not containing the quoted numbers, no written endpoint convention. §10, not deleted.

2.3 What we explicitly do ****not**** claim (shared tooling)

The companion and this paper share project-internal process protocols — a multi-model review checklist and a local/cloud scripting discipline — and neither claims originality for them:

Supplementary S12, *Shared process protocols*.

3. Units: the methodological premise

"Have a split problem" is a property of a (**benchmark, counting unit**) pair, not of a benchmark. We use four units, defined by **who controls the split**:

- **`release`** — the split the benchmark's **own published release** ships: the `train: / val: /` test: entries and paths in the distributed archive. *Example*: an archive whose `data.yaml` binds `val:` and `test:` to the **same directory** records "yes" here (§4).
- **`protocol`** — the split the benchmark's **official protocol** tells you to use, which need not be the released one: a protocol may carve its own validation set out of the released training split. *Example*: an aliasing release whose documented protocol still requires a held-out test split records "no" here and "yes" above — the units answer different questions.
- **`yolo\_dist`** — the split a **generic YOLO distribution package's** own configuration resolves to, i.e. what a practitioner taking the pipeline off the shelf gets. *Example*: a clean release whose community YAML points `val:` and `test:` at one directory records "yes" **only** here.
- **`reported`** — the split the number **the literature prints** came from (not the official leaderboard's

own score), read from the release documentation and the dataset's own paper. *Example*: two papers on one benchmark can

differ here while agreeing under all three units above — hence the headline count uses this unit (§6.1).

A count is meaningless without its unit: the same nineteen benchmarks answer differently under each

(Fig. 1), so every count names its unit.

The proposition is not ours to claim. That **the same data yield different defensible audit counts under*

*different operational definitions** has been stated independently, for molecular benchmark auditing, in a 2026

preprint whose abstract closes "different defensible definitions return different answers from the same

file" [18]. **The observation is the cited literature's; the instantiation, and the four named units, are ours.** Our contribution is therefore ***not the observation but its instantiation in detection*

evaluation*: *naming the four units by* who controls the split*, and backing them with the 19-row, four-unit

evidence table. The decomposition is also orthogonal to [17], which separates leakage by **type**: we

separate the **counting subject**, so one leakage type can still be counted under four units (§6.1).

Table S2 → Supplementary §S8.

The 'reported' unit is read from the release documentation and from the dataset's own paper: we did not sample papers per benchmark and make no claim about prevalence; a stated-frame sample is future work (§11). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check — the full 19×4

table and the cross-check against the original audit are **Supplementary S1**, the derived counts §6.1 and its

table.

The rule itself is not new. The same toolchain's own course states it normatively — *"Test set is touched once, at the end. Not for tuning." and "The splits must be independent on the dimension that*

*matters at deployment." [19] — so what this paper adds is not the rule but the audit and the count**

under it.

The premise we rely on throughout: a "yes" under any unit means *the reported number is not a sample from an independent split*. **What that does and does not license is stated where the counts are printed** (§6.1 and the note under its table).

3.1 Formalisation: what the four quantities are, and what they can and cannot identify

Setup. Training is a randomised map (seed ω , recipe θ , checkpoint e) $\mapsto$ weights; evaluation a fixed functional M_S on a labelled split S , with $V(e) = M_{\{S_{\text{val}}\}}(W(e))$, $T(e) = M_{\{S_{\text{test}}\}}(W(e))$, $\hat{e} = \operatorname{argmax}_e V(e)$. Sections 5–9 measure four quantities: the selection premium, Δ_{gap} , κ , the realization rate.

Definition — the raw gap is not identifiable from the pair alone. $G = V(\hat{e}) - T(\hat{e}) = \delta + o$ mixes a split-difficulty term with a selection-optimism term; the decomposition, **and the reading that a single-model endpoint pair is unable to separate its two terms, are [20]'s**, while its **strong form** — that *any* δ' fits the observation equally well — is **stated here as a formalisation rather than attributed to that literature. The pairing assumption is what makes Δ_{gap} meaningful**, failing by construction under aliasing so that §9 cannot be read as a clean-protocol

result; the identity holds value by value on all 13 scanned cells and both escalated cells at $n = 10$ (§5.6).

Adopted identity, not a claim of this paper. With $V(e) = v(e) + \varepsilon_e$ and $T(e) = v(e) + \tau + \eta_e$, $E[\text{premium_test}] = E[\text{premium_val}] - WC$, $WC := E[\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$, so the realization rate is $1 - WC / E[\text{premium_val}] \leq 1$, with equality only if selection is driven by signal alone [16] (**the per-run check is on the observed 'prem_val - prem_test', so it corroborates that consequence and is not an independent test of the latent 'E[$\varepsilon_{\hat{e}} - \varepsilon_{e_f}] \geq 0$ ', which no archive settles without the identifying assumption**) — the optimizer's curse in this paper's notation, the frame for §8.6–§8.7. **Two consequences are testable on the archive** — the per-run inequality, and monotonicity in the val noise (**Supplementary S11**). A third form, that the premium scales as $\sigma\sqrt{(2 \ln E)}$, was **tested and withdrawn (Supplementary S11)**; contemporaneous work measures $\approx 5\times$ on the same $\sigma\sqrt{(2 \ln K)}$ bound [21].

Corollary (the premium is mostly shape, and the peak is the worst place to stand).

$\text{premium_r} = \text{shape}(E) + v_r$ holds **exactly**, $\text{shape}(E)$ being the mean curve's peak-to-final gap, so shape is independent of σ and $E[v_r] \geq 0$. On our 40-run four-budget series **the monotone criterion is met on the clean re-run** while the $\sigma\sqrt{(2 \ln E)}$ form is **outside** its pre-declared band — so that scaling **stays withdrawn** — and the pre-declared shape control passes on the mean it specifies while failing per arm; **both readings are reported, and the with-early-stopping series carries the pre-declared verdict. We therefore do not claim an 'E' effect**, and the checkpoint rule is stated as a **reporting rule, not a candidate** (the top-5 mean beats the argmax on the same 40 runs), with **three limits** — an empirical capture rate, a val-side cost the argmax does not carry, and a two-split, five-epoch-grid scope. **Every per-run value, both readings, the regression and the limits are in Supplementary S11.**

What is ours, and what is not. Of the statements that follow, the raw gap's definition [20], the impossibility result for endpoint-pair estimators [22] and the family-dependence procedure [23] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**. Full statements and proofs are in **Supplementary S11**, and the detail displaced from here in **Supplementary S12**.

3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first

The propositions of §S11 are testable, so we ran them on rented GPUs with the **criterion fixed before the runs; Supplementary Table S37** gives, for each of the six experiments, the criterion it was judged against. Counts below are over **experiments**, not criteria.

The two negative outcomes and the redone design are stated as such in Supplementary Table S37 (G1 is void, and **two of the five judged experiments are negative** — G1' and G2); the values behind every verdict are in **Supplementary S31–S33** (the G1 family in S12).

The reading below is stated more weakly in **Supplementary S11, section 13.3**.

We then ran the clean test of G3's epoch-budget series **on one machine with early stopping disabled**, all four budgets present: C1 is **met**, C2 falls **outside** its band, and C3 **passes** on the

mean $\hat{e}/E$ it specifies but not per arm — see §3.1 and Table S34b. The pre-declared verdict stays with the with-early-stopping series; the clean re-run is the reading used here. The judged criteria carry one message: **the selection bias moves with the absolute amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file**. Two consequences do not follow: the configuration-level fact of §4 stands — the reported number *is* the selected number — while the stronger reading, that aliasing *adds* a measurable amount on top of it, does not.

On the shared 5-epoch grid the val and test peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs: where they agree the **median Δ gap is lower** (mean **+0.13 pp**, median **−0.20 pp**, $n = 7$; **post-hoc**), where they differ it is **systematically not** (mean **+0.91 pp**, $n = 33$; permutation $p = 0.009$, **post-hoc**; the median reading is $p = 0.0011$). **This is a partial predictor, not an explanation of the sign**; its limits — including the four of seven agreement runs whose gap is non-positive — the per-run table and the test are in **Supplementary S11**.

4. Finding 1: our own corpora

4.1 How common the aliasing is

In **our own** experimental corpora, **6 of 14** give the reported number no independent held-out split — **AI-TOD, DOTA v1, DOTA v1.5, Mendeley face-mask, SFCHD and SHWD**; five of the six bind val: and test: to one directory, and **SHWD is the sixth, where it is `train` and `val` that share one** (**Supplementary S1**, row 13); in **4** of the remaining 8 the two keys are genuinely distinct, and the last **4** have no test: key at all (a different failure mode), so the three groups account for all 14 — **a count over the corpora we happen to hold, not a prevalence**. Counting by file, not by corpus, **19 YAML files** alias the two keys, in five locations; on the archived manifest subset, which covers only part of these corpora, the ratio is **8 of 28 archived configs**. **One corpus aliases a different pair: SHWD's distribution configuration binds `train` and `val` to one directory** — the release itself ships no YAML, and the binding is read from the mirrored community configuration that **Supplementary S1**, row 13 records (evidence class A, medium confidence).

For the corpora whose `val` and `test` are the same directory this is not contamination: their image-level intersection with training is **exactly zero**, so the aliasing changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question. **SHWD is the exception and is not folded into that statement**: with train and val on one directory, its training and evaluation images are the same files. **It still counts among the six, for the reason the list is about: its `val` is not an independent held-out split either**.

4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [24]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [25]). §10's first recommendation: make "are val and test the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

5. Finding 2: the selection premium — magnitude, direction, and mechanism

Evidence. `eval_validity/selection_premium_local_20260916.txt`, reading **569 archived runs** from the local archive (the local version is the 2026-09-16 repair).

Definition. `premium = best-epoch – final-epoch mAP50-95 within a single run`. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [26] (early stopping, data order) and [27] (benchmark variance across seeds). Being a maximum over a noisy curve, it also carries the upward bias of any selected extreme [16]: what those works average over as uncertainty, we measure as a reproducible component. **The two arms.** Every cell is a paired comparison in which the **baseline** arm follows the published convention and the **strategy** arm is the intervention under test — in the cells reported here, raising the peak learning rate from **0.001 to 0.005** with everything else held fixed (batch 32, `imgsz 640`, SGD; YOLOv12n with the shape-aware IoU). The two arms run the same code and the same data.

Table S3 → Supplementary §S8.

5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: "the convention understates the effect" is wrong, and external checking refuted it (§5.4, II). **Endpoint convention (frozen):** `best – last within a single run`, both endpoints reported for every cell; §7.2–§7.3 quantify the other convention.

Tier 1: the selection premium is positive in all four cells and the baseline arm gains more, consistent with the selection-bias literature, not new in kind; **on the monitored split that positivity is near-definitional** (best is that curve's own argmax), and **the finding we report is the paired arm difference and its test side** (§8.6). Pairing brings **no** variance benefit:

the paired *t* of 5.74 and the unpaired Welch *t* [28] of **5.68** agree — at $p = 0$, ****standard arithmetic** rather than an empirical finding of ours [27]. Permutation checks. **A sign-flip test on the same paired differences agrees with the *t*-tests at each cell's attainable floor** (Supplementary S12**).

Meta-analyses of published results have long quantified this optimism: development-set early stopping

is measured to *overestimate realistic model performance*, by as much as **18.0 %** accuracy [29]; and a

large Kaggle meta-analysis finds *little evidence of substantial overfitting* [30] — complementary rather

than contrary, since its mechanism is holdout **reuse** and ours needs none.

Metric sensitivity. On `metrics/mAP50(B)` the mean direction holds in all four cells, but

`smoke2sf` loses both its significance (**+0.063 pp**, $p = 0.54$) and its sign unanimity (**6+/5-**, not

11+/0-): **the mean direction is metric-robust, one cell's significance and sign unanimity are not**

(full mAP50 columns and both mAP75 columns — the mAP75 `val` side is a **5-epoch grid**, so a rate computed from it is **upward-biased** — **Supplementary S12**).

Table 1. Tier 1 — each arm's **peak-final gap** and the **paired difference** between arms, by cell; the gap is a within-run maximum (§5.1), so the finding is the paired difference. n counts paired units (ten seeded, one unseeded — **Supplementary S12**); at $n = 10$ the two differences hold (**+0.272, +0.449 pp**). **A different run set from the transfer measurement in §8.6:** these are the 569-run local archive under the published (aliased) protocol, while that measurement's premiums come from the clean three-way runs — the two tables' premiums are different quantities, not two readings of one. Source: `verify_two_critiques_20260916.txt`; the `dota15` row's $n = 10$ escalation is in **Table S28**.

Cell	n (paired units)	peak-final gap, base	peak-final gap, strat	paired diff	sd	t	p	95 % CI
---	---	---	---	---	---	---	---	---
smoke2sf (strong)	11	+0.799	+0.382	+0.416	0.240	5.74	1.9×10^{-4}	[+0.255, +0.578]
shwd2sf (corroborating)	11	+1.038	+0.512	+0.526	0.207	8.41	7.6×10^{-6}	[+0.386, +0.665]
a2d15 (directional probe)	10	+0.534	+0.262	+0.272	0.183	4.69	1.1×10^{-3}	[+0.141, +0.403]
dota15 (within-domain)	10	+0.941	+0.492	+0.449	0.196	7.23	4.9×10^{-5}	[+0.309, +0.589]

Table 2. Tier 2 — the paired difference Δgap by corpus. $\Delta\text{gap} > 0$ means `val` understates the strategy arm's effect. **Δgap is a split-dependent arm contrast, not an isolated selection effect:** it decomposes into a difficulty gap and a selection term, and **Δgap 's sign is not the selection term's sign** — on the two headline cells the difficulty term dominates while the selection term is small or opposite-signed (**Fig. S1**). That the two arms see equal split difficulty is an **assumption we state**, not something this design demonstrates; the selection contribution proper is $\Delta\text{prem}_{\text{val}} - \Delta\text{prem}_{\text{test}}$. Source: `xeval_analysis_20260916.txt`.

Corpus	n	Δgap (pp)	p	95 % CI	Reading
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shwd2sf	10	+0.779	0.0053	[+0.297, +1.261]	val understates
smoke2sf	10	+0.461	0.0263	[+0.068, +0.855]	val understates
p_aitovis	10	-1.226	1.3×10^{-8}	[-1.370, -1.081]	val overstates
p_vistod15	10	-3.315	6.5×10^{-9}	[-3.676, -2.953]	val overstates

The split of Δgap is recorded for two of the four cells: difficulty **+0.759** / selection **+0.020** (shwd2sf), and difficulty **+1.041** / selection **-0.580** (smoke2sf) — in the second the selection term is **opposite in sign to Δgap** , so the sign does not carry it, and **both intervals include zero** (Supplementary S12, Fig. S1). The two p_ cells' prem_test was **not recorded at that time**: their per-epoch test curves — and with them prem_test and WC — were evaluated on 2026-09-30, so nothing is imputed and the cell-level values are now recorded (Supplementary S12, Table S38).

The four rows below are an escalated base, not a sample: the two p_ cells reached $n = 10$ because their $n = 3$ Δgap had been flagged, so the 2/2 balance is a property of the escalation rule; the other nine cells are **descriptive** (Supplementary S12). This cell is also the one where **3 runs** changed training path silently under GPU memory pressure, and why we do not call it clean is in §10, defect 10.

(Third §5 table, S5 → Supplementary §S8.)

Tier 2: on the same three-way corpora the paired difference Δgap varies in sign. The raw $M_{\text{val}}(\hat{e}) - M_{\text{test}}(\hat{e})$ is **not identifiable**; the paired difference cancels difficulty —

$\Delta\text{gap} > 0 \Rightarrow$ 'val' understates. Its effective base is four cells, not thirteen, and an escalated base rather than a random sample, for the reason Table 2's note gives; “2 understate / 2 overstate” is therefore a statement about these four cells, not a rate for the family. No prospective power analysis was registered; the scan is in Supplementary S12.

Both escalations met the criteria fixed in advance. dota15 **+0.449 pp** (10+/0-);

p_aitovis **-1.226 pp** and p_vistod15 **-3.315 pp** (0+/10-), permutation $p = 0.001953$,

the attainable floor at $n = 10$. The best.pt endpoint was identified, not assumed: on test it reproduces the archived sio_b_results.csv to **0.0046 pp**.

The formal claim: the convention systematically shifts **arm-to-arm comparisons** (Tier 1: four cells, CIs excluding zero); **the direction of that shift varies by corpus** (Tier 2: 2 understate / 2 overstate on the four-cell base) — **neither "a general understatement" nor "no systematic direction"**.

BH at $q = 0.05$ **now leaves three cells significant** — p_vistod15, p_aitovis, shwd2sf [31],
****reversing what we reported before the**

escalation**.

Scope (cross-framework boundary). The convention this paper audits belongs to the **YOLO lineage —

Ultralytics YOLO [19] and YOLOX [32]**: at pinned commits, Detectron2 (`TEST.EVAL_PERIOD = 0`) [33],

MMDetection [34], DETR [35] and PaddleDetection [36] **report the final epoch instead** (§6.5) — that

pipeline family's convention, **not the field's**.

5.2 Two versions must be disclosed side by side

Two archives, two versions of this table. The main text uses the **local archive** (569 runs, $n = 11$); the pod root (347 runs, $n = 10$) is a **Supplementary S12** cross-check agreeing in sign and magnitude, not in value.

5.3 The dota15 cell: now reproducible, and one wording defect exposed

This cell was not reproducible from the release as first written, and the repair exposed a wording defect (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

Supplementary S12 carries the two external critiques in full. **We used to write** that the gap was simply unidentifiable; the

external check **corrects this to** the paired-difference form, which cancels the difficulty term

under the pairing assumption stated in §S11. The argument and the recomputation are in

Supplementary S12.

5.5 The same component in third-party public logs

The same quantity appears in other people's logs: 20 logs from 6 sources, median **+1.04 pp**, max **+5.65 pp**, 18/20 positive — a sample, not a field-level rate. Found by repository **search** (Kaggle and public repositories); the queries, the rest of the frame and the reason public logs **cannot establish aliasing** at the configuration level are in **Supplementary S12**.

Five limitations: ① the two cells escalated in §5.1 reach $n = 10$ with **local** runs and test-side evaluations, and the remaining 9 cells' test side was closed on 2026-09-30 on one machine

(**Supplementary S12**, Table S38), but their runs are still **not local** to the headline cells' host;

② `prem_test` is reported as **best – last**: the per-epoch test curve now exists for **all thirteen**

corpora, but it is not what this rate uses; ③ `last.pt`'s val recomputation differs from the final `results.csv`

row by ~ 0.02 pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — `prem_val` across the five arm-cells spans **+0.605 to +1.469 pp** (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points, and the aggregate summarizes a wide spread rather than a typical run's outcome.

5.6 Why Δgap 's sign varies by corpus: three of five candidate mechanisms are refuted by the data

Three of five candidate mechanisms are refuted, and the mechanism itself is not identifiable — the sign of Δgap is a sum of same-order terms, so no single mechanism can be read off it (§S11, Remark 3) — **but the paired quantity is, and is what we report**. The examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–S10.

6. Finding 3: how general this is across public benchmarks

Evidence: `benchmark_split_audit.md`, a line-by-line audit of 19 benchmark rows, plus

`split_units_19rows_20260916.md` (recomputation script; no hard-coded percentages).

The 19 audited benchmarks, with the release or dataset paper read for each: COCO [37], PASCAL VOC [24], Objects365 [38],

Open Images v7 [39], DOTA v1.0 and v2.0 [40], VisDrone-DET [41], AI-TOD [2], UAVDT [3], xView [42],

DIOR [43], NWPU VHR-10 [44], SHWD [45], SFCHD [46], MAFA [47], Mendeley face-mask [25], WIDER FACE [48],

CrowdHuman [49], D-Fire [4]. **These nineteen are the configurations this archive could read, not a sample of the field.** The question is **evaluation validity**. Dataset-level bias auditing [50]

and classifier-accuracy estimation [51] are neighbours: both ask whether a reported number measures what it

is taken to measure, but neither audits a release's configuration binding.

6.1 The result under four units (single-rater judgements)

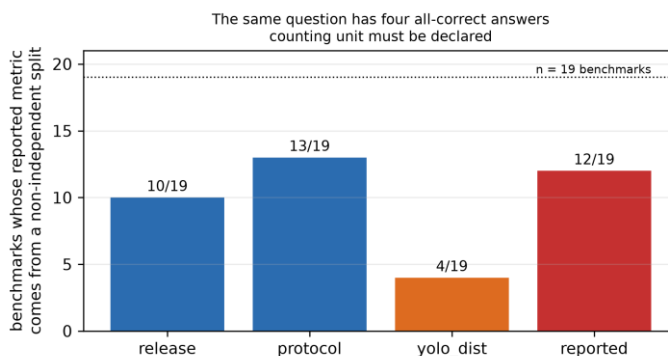


Table 3. The four counting units over the 19 audited benchmarks. **A count is meaningless without its unit (§3).** Source: split_units_19rows_20260916.md. Rows marked `n\_a` enter the denominator, not the numerator, and are not undecidable; the other readings are 12/19 for release and 10/15 $\approx 67\%$ for yolo_dist (six rows enter that denominator only), and 12/15 is the 19 rows less the four n_a rows; under the COCO-val reading the reported unit is 13/19.

Unit	non-independent	of decidable rows	undecidable
---	---	---	---
release (the benchmark's own release)	10/19 $\approx 53\%$	10/19	—
protocol (the official protocol)	13/19 $\approx 68\%$	13/19	—
yolo_dist (the generic YOLO distribution package)	4/19 $\approx 21\%$	4/15 $\approx 27\%$	4 rows
reported (the layer actually reported)	12–14 of the 19 $\approx 63\text{--}74\%$	12/15 = 80 %	4 rows
↳ of which literal two-key same path (= the 2/19 the original audit reported)	2/19 $\approx 11\%$	2/19	—
reported – clean	3/19 $\approx 16\%$	3/15	4 rows

What a count here is, and what a "yes" means. Every row is a **single rater's judgement**

(§6.1), so the counts are printed as counts — the percentages beside them are a reading aid, not a

rate estimated from independent coding. A "yes" under any

unit means the reported number is **not a sample from an independent split**; it does **not** mean

the benchmark is broken or the experiment dishonest, and it does not license a generalization claim.

These 19 rows were coded by one rater — hand-written, 28 literal judgements (§10, defect 5) — and **eleven blind language-model coders across four rounds** reached only **fair-to-moderate** agreement with them (per-coder κ 0.11–0.44, weakest on reported throughout); a **non-blind, model-assisted** pass agrees at κ 0.874 (Supplementary S12) and **moves two rows**. reported means **the number the literature prints, not the official leaderboard's**, so it is printed as a range, 12–14 of the 19 $\approx 63\text{--}74\%$ (the 80 % column is over the 12/15 decidable rows); rows marked n_a enter the 19 but not the numerator (Supplementary S12).

Five clauses move to Supplementary S12 (the self-check's column and subtotal equalities to **S1**): the COCO row's judged number, the yolo_dist first-version correction (**4/19**), the literal-two-key self-check (**2/19**), Mendeley face-mask's **generator** provenance, and the audit's unused "63–79 %" bound.

6.2 "Same directory" has four levels, and the audit reported only the narrowest

The levels, the rows each admits and the resulting shift are in **Supplementary S12**, Table S12.

6.3 We do ****not**** reproduce the original audit's two numbers, and we say why

one rests on the narrowest reading of "same directory", the other on a stale snapshot; both recomputations are in **Supplementary S12**, Table S13.

6.4 One judgment the evidence does not support (corrected)

One classification correction is recorded in Supplementary S12 (D-Fire moves to the clean group).

6.5 Across frameworks: is the convention specific to YOLO?

Among the pinned default configurations we checked, only the YOLO lineage does this by default. At pinned commits Detectron2 (TEST.EVAL_PERIOD = 0), MMDetection, DETR and PaddleDetection **report the final epoch instead** (file-by-file evidence, 25 configuration files: **S12**, Table S14; **a targeted sample at pinned commits, not a census**, so "the convention" means that family's, not the field's). A **cross-lineage realization test** — a Faster

R-CNN (ResNet-50 FPN) detector, a family that **reports the final epoch by default** — on VisDrone under the

same three-way split, at **six seeds**, is **positive in all six runs, with a 95 % interval excluding zero** (**68.2 %** realized; **+0.0809 pp** mean prem_val, 95 % CI [**26.9 %**, **123.5 %**], per-run **2.8 %–174.8 %**)

(values, interval and per-run table: **Supplementary S34**): the mechanism is **not confined to the YOLO lineage in this one instance**, which we read as an **illustrative case rather than a cross-lineage result**: one detector, one corpus, six seeds, and a magnitude that is not comparable across detectors.

7. Finding 4: what "significant" means, and how the endpoint convention changes it

7.1 With $n = 3$ a permutation test ****cannot**** reject in principle

A power note for the cells that remain at $n = 3$, with its derivation. Recovering each Tier-2 cell's sd from its own printed 95 % CI ($sd = \text{half-width} \div (t_{\{9,0.975\}}/\sqrt{10})$) gives **0.20–0.67 pp** across the four escalated cells; a paired comparison at $n = 3$ then needs $(t_{\{2,0.975\}} +$

$t_{\{2,0.80\}} \cdot sd/\sqrt{3} = 3.10 \cdot sd$, i.e. **0.6–2.1 pp**, for 80 % power at a two-sided $\alpha = 0.05$ — **at or above every Tier-1 effect this paper reports (+0.27 to +0.53 pp); at $n = 10$ the same derivation gives an MDE of $0.99 \cdot sd$, i.e. 0.2–0.7 pp**, the scale at which the four escalated cells are judged. The nine cells left at $n = 3$ are therefore not merely unlucky: **at this design they cannot carry an inference**, and they are reported descriptively only.

For a sign-flip permutation test on n pairs the smallest attainable two-sided p is $2/2^n$: at

$n = 3$ that is **0.25**, so such a cell has **no resolution at this n** , and ******after §5.1's escalations

no headline cell (Tables 1–2) is $n = 3$. The derivation, the floor at every n and the reading each n

licenses are in Supplementary S12.**

7.2 Both conventions become significant only at $n \geq 6$

Both conventions become significant only at $n \geq 6$ — below that a permutation test has no resolution (§7.1). The clean-protocol seed extension that reaches $n = 10$, and the per-cell numbers, are in **Supplementary S12** and Table S15.

First-report attribution (a convention we hold to). Three things here belong to the companion

paper and are cited, not claimed: the two Δ values (**+1.434 / +0.507 pp**), the three-way remediation

protocol itself, and the pre-registered multi-target replication (§9). What this paper reports first is

******the selection premium (§5), the 2×2 realization of that premium on a disjoint split (§8.6–8.7), and

the formalisation of the realization rate (§3.1)**. The full statement — including how the companion is

named to the editor while it carries no identifier — is in **Supplementary S12**.

7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

One run is recorded with two different endpoints in two archives — which checkpoint is recorded was never specified: **Supplementary S12**, Table S16. taking the final-epoch row (67.12) gives $\Delta = -1.490$, $p = 0.0141$; taking the best-epoch row (70.06) gives $\Delta = -0.902$, $p = 0.2931$.

The headline tables of this article report the 'best' reading — the frozen endpoint convention of §5 — so that cell is **not** significant as printed; the final reading appears only where §7.2–§7.3 compare the two conventions. The affected run, named with both archived endpoints in **Supplementary S12**, appears in the article only in the `sio_b` cross-check and §10's defect table; the ambiguity therefore **does not move a number printed here**.

8. Finding 5: how far the selection premium transfers to an independent split

8.1 The protocol is the companion's; the measurement under it is this paper's

The three-way remediation protocol is **defined and reported in the companion paper**, as a first report

there (the finding that **the effect reproduces in direction while its magnitude falls**), and this section

therefore **cites it** rather than presenting the replication. ****In this paper's own terms it is three**

things******: a val carved from the pool so that training never sees it; the target corpus's own held-out

test, with a filename-level check that its images do not intersect training; and an endpoint ****defined in**

writing before any run. **The registration text and its FROZEN-HASH**** stamp are reprinted in

§9 and Tables S24–S27 and the frozen glossary, so this package carries them. What this paper measures

under that protocol is the quantity §3.1 formalises: ****how much of the val-selected premium survives**

on a split no selection rule can touch******, and that measurement is this paper's.

8.2 The protocol's results on the companion's corpora: ****transferred****

The protocol's results on the companion's corpora are the companion's first report. What this paper adds under the protocol is the measurement of §8.6–§8.7; the transfer (2026-09-21) is recorded in the module-transfer note of that date.

8.3 A direct measurement of selection inflation

A direct measurement of data-order replication — reproduced inside this setting: adding replicates that permute the training file order lowers the mean by **9 %–29 %**, each figure taken **against that cell's own n = 6 mean** and not against the pooled

headline effect, the **seed-budget discipline** measured rather than assumed (the setting in which [52] locates the first-order component, §8.4). **What these replicates are, and what the bound is a bound on**. The per-run randomness the ten runs install is a **deterministic permutation of the training file order** (`--shuffle-seed`, the number in the run's own name), **not** the framework's seed argument: the Ultralytics data pipeline does not draw from `args.seed`, and on two of these corpora different `--seed` values produced **identical** epoch-by-epoch replays (**Supplementary S12**, section 9.3). The quantity is therefore **data-order replication**, not seed variance: the 9 %–29 % it bounds is the **file-order** component of the reported mean, and is **not** an estimate of the

variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** — §8.6 positions the closest concurrent work, which measures the **dose-response** over 2,047 datasets — so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards**; design and the check against recomputed means: **Supplementary S12**, Table S18. **The title is deliberately the weaker, sample-scoped claim**: with three initializations the interval is wide, so the components are **consistent with, not established by**, these three initializations.

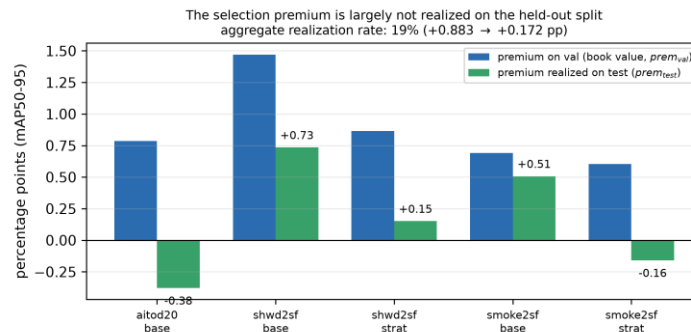
8.4 The three variance components: the excluded one is the smaller in this sample

The three variance components, and the excluded one is the smaller in this sample: the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order — the **same ranking** as the shuffle-order result of [52] and consistent with the run-to-run variance literature [26]; we therefore **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is a sample statement, is in **Supplementary S12** and Table S19.

8.5 Why "same-pipeline pairing" is a necessity, not a habit

Mixing two pipelines' val/test conventions would confound the split convention with the framework. The reasoning and the counterexample are in **Supplementary S12**. **Pairing is not neutrality, however**: the paired design used here covers the **budget and data axes only**, and **not** the optimiser's.

8.6 The premium's realization rate: arm-cell median ****17.6 %**** (five arm-cells, $n = 10$; run-weighted ****23.3 %****)



What this section claims, and what it does not. The rate below is the realization of the val-selected premium on a disjoint split ****for the five arm-cells measured here, under the companion's protocol** — not a constant of the convention. **The escalated cells reach a pooled 55 %, but the two batches are not pooled**** (different runs, different protocol, §8.1). "The convention realizes about a quarter of the val premium" would be false; "in these five cells, about

a quarter survives" is what we measured.

How much of the gain selected on `val` survives on `test`?

Table 4. The premium selected on `val` versus what that same selection action realizes on the disjoint `test` split, per arm-cell (**an unbalanced pool** — two corpora contribute both arms, `aitod20` only its baseline — so the arm-cell is the unit). Source: the **per-epoch `val` curves** (`results.csv`, for `prem_val`), `xeval_20260916/matrix.csv`, and this

revision's two batches (`x4_teval.csv` on A, `x4fill_20260925.csv` on B, for `prem_test`). **The §10 memory-pressure bound (defect 10) does not arise in this table:** its hits are in §3.1's clean batch and §5.1's `p_vistod15`, none of the five arm-cells below.

Cell	Arm	n	<code>prem_val</code>	<code>`prem_test`</code>	realization rate
---	---	---	---	---	---
<code>shwd2sf</code>	baseline	10	+1.469	+0.736	50 %
<code>shwd2sf</code>	strategy	10	+0.865	+0.152	18 %
<code>smoke2sf</code>	baseline	10	+0.691	+0.509	74 %
<code>smoke2sf</code>	strategy	10	+0.605	-0.157	-26 %
<code>aitod20</code>	baseline	10	+0.766	-0.218	-28 %

where `prem_val` = $V_{\max} - V_{\text{final}}$ (**selecting on val**) and

`prem_test` = $T_{\text{best}} - T_{\text{final}}$ (**realized on test**).

Aggregate (runs weighted equally, $n = 50$; the `aitod20` cell completed from 1 run to 10): ``prem_val` +0.879 pp` → ``prem_test` +0.205 pp` — arm-cell median 17.6 %, span -28.5 % to +73.7 % (102.2 pp), 2/5 negative; run-weighted, a realization rate of 23.3 %, reported with the cell-level interval as the primary one: [-17.3 %, 53.2 %] (**five clusters, $P(\text{rate} \leq 0) = 0.118$**), the run-level bootstrap giving [8.4 %, 37.4 %] ($P(\text{rate} \leq 0) = 0.002$) as the narrower sensitivity (script ``work/item2_final_v2_20260925.py``). On a difference scale the eleven clusters give ``prem_test` - prem_val` = -1.367 pp`, 95 % CI [-1.59, -1.15] (not a new experiment; **Supplementary S12**), on a denominator that spans +0.605 to +1.469 pp across the five arm-cells. The same re-aggregation's arm-cell-equal rate is **36.9 %**, 95 % CI [19.3 %, 60.4 %] — the interval's sign depends on the base. **Completing the cell, not the weighting, removes the spread** they showed on the previous 41-run sample. **Leave-one-out 9.77 % to 34.16 %:** *partial realization* is a **descriptive five-cell statement, not significant** at claim level; in two of the five arm-cells the selection action does not merely fail to pay off, it **costs** — the `smoke2sf` strategy arm and the `aitod20` baseline. Closest concurrent work (**82 % / 104 %** and **56 % / 29 %**, pooled **55 %**), the winner's-curse check, the withdrawn $\sigma\sqrt{(2 \ln E)}$ form, and each arm-cell's corpus pair and public status: **Supplementary S12**.

(the best.pt-versus-argmax residual, our withdrawn attribution and why the remainder is **unidentified: Supplementary S12.**)

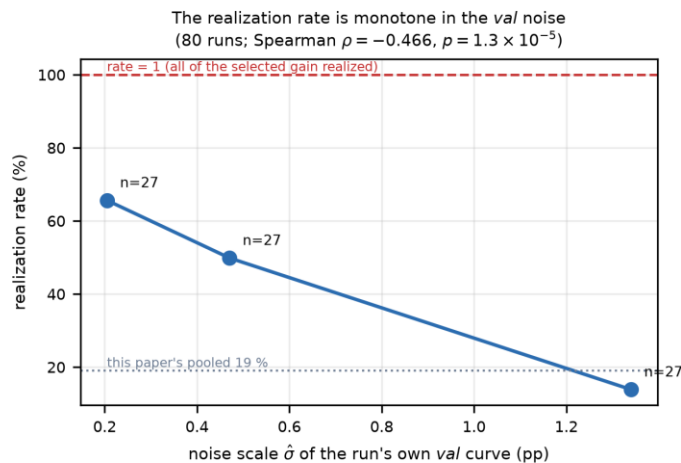
The monotonicity is a correlation, not a causal claim, and at the arm-cell unit the nine-cluster reading is $\rho = -0.433$ ($p = 0.244$) — see **Fig. 4** and **Supplementary S12**: the $n = 80$ are the 81 runs with a measured

test side less the one whose `prem_val` is 0. It has a named precedent: [53] gives validation overfitting (VO) its name and bounds the expected best observed loss by

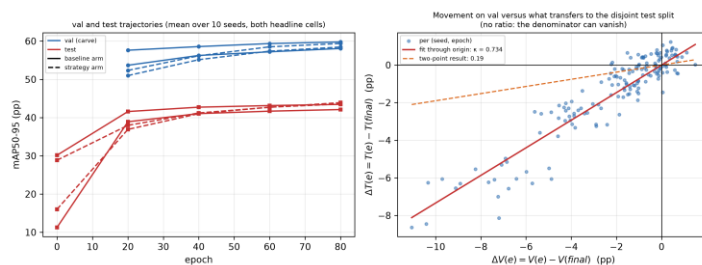
$E[\min L] \geq v^* - s_m \sqrt{2 \log(Nq)}$, an optimism growing with the number of selection queries Nq .

Recomputation status. The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; §10, defect 5); a GPU 2×2 **evaluation** on **41** three-way runs of **YOLOv12n** [54] $\times$ {best, last} $\times$ {val, test} makes them so, **retraining not**, reproducing the archived best $\times$ test values to within **0.0047 pp** (median **0.0021 pp**) — the endpoint rule's positive control (**Supplementary S12**).

Three limitations: ① in the **41-run epoch sweep** only the two headline cells reach $n = 10$; `aitod20` contributes one run of 30 epochs and 6 checkpoints where others contribute **20** (hence 806, not 820), its Table 4 arm-cell a separate $n = 10$ batch; ② **closed: every retained checkpoint is now evaluated**, the *stability* reading is narrower than the grid; ③ κ is relative to each run's own final epoch, not causal. The 806-row evaluation is archived as `xeval_perepoch_20260918/matrix_perepoch.csv` with a SHA-256 list.



8.7 The per-epoch view: the realization rate is not a constant



One metric self-correction first. Our first version used $r(e) = \Delta T(e) / \Delta V(e)$, producing values such as **−860 %** and **+1031 %**: at late epochs $V(e) \approx V(\text{final})$, the **denominator vanishing** — **an ill-conditioned metric, not a finding**. We report only the non-dividing $\Delta V(e) = V(e) -$

$V(\text{final})$ and $\Delta T(e) = T(e) - T(\text{final})$, summarized by the through-origin slope κ of ΔT on ΔV .

Tables S21–S23 → Supplementary §S8.

Slope κ (how much of the movement on val transfers to test). On the **complete curve** —

$n = 765$ (seed, epoch) points over the 41 runs — $\kappa = 0.708$, $R^2 0.937$, with a run-clustered

(sandwich) 95 % CI [**0.671, 0.745**]; the within-run estimator gives **0.698**. Across the curve ΔV reaches

−8.1 pp and about **73 %** transfers: **early-training progress is real progress**. §8.6's realized

share measures the **selection-relevant sliver** near the peak, where ΔV is of order **± 0.2 pp** — so

******"most of the movement on val transfers" and "the little extra gained by selecting near the peak does

not" are both true, **and the premium is the residual of the transfer rate**. And κ is not constant

across arms either — in the direction the selection account predicts: baseline $\kappa = 0.600^{**}$

([0.581, 0.619]) against **strategy $\kappa = 0.767$** ([0.735, 0.800]), difference

+0.168, 95 % CI [+0.130, +0.203].

A second result, and a self-correction the curve forces: the sign-of- Δgap claim we wrote from four

sampled epochs is **too strong for one of the two cells** (on the complete curve smoke2sf starts

negative) and is **narrowed here**; per-checkpoint values: **Supplementary S11**.

The limitations of this section are in Supplementary S12.

9. The registered multi-target replication: ****transferred to the companion paper****

This paper does not restate its numbers **as claims of this paper**: the registered endpoint table is reprinted as the **companion's** in **Supplementary S24–S26**, and this section gives only what a reader needs here to follow the argument.

The pre-registered multi-target replication is **reported in the companion paper**, where it is a **first report**: its frozen registration (**2026-09-13 01:37:27 UTC, FROZEN-HASH md5 6a7eee7b3e34b15ce5adcba14cf7ea36, zero runs existing at freeze time**) and its verdict ("**criterion not met**", written deliberately **not** as "hypothesis refuted"). This section states the

protocol points a reader needs in order to read that registration from inside this paper; the endpoint table above is reprinted as the companion's, and none of its results is stated here as a claim of this paper.

Data. Three registered target corpora, each taken from the public pools this paper audits and each carrying its own train / val / test configuration; the test half is the corpus's own held-out split and no selection rule sees it. Every run is a directory whose name carries its corpus, its arm and its --shuffle-seed; the per-seed table for **all 90 runs** is deposited with the registration.

Splits and the two arms. The registered pairs ran under the protocol in force at registration, which is the corpus's own train / val / test configuration — and there the **'val' side aliases the 'test' half**, the alias whose bias term Table S27's note quantifies ($\Delta \approx \Delta_{\text{reported}} + \bar{d}$). The three-way carve, with the carve taken **out of the pool** and all three pairwise intersections **verified zero by filename**, is the protocol §8.1's own clean batches use; the two are not the same design. The **baseline** arm is the published convention — the checkpoint the pipeline's own fitness rule keeps (best.pt) at the peak learning rate **0.001**. The **strategy** arm is the registered intervention: the shape-aware IoU (shapeiou) at **0.005**, everything else held fixed (batch 32, imgsz 640, SGD, YOLOv12n). Both arms run the same code on the same data, and the endpoint definition is frozen in writing as §8.1 states.

Criterion status. The criterion was fixed **before any run existed**: the intervention had to move the designated cell in the registered direction, at the registered significance and with the registered sign split, and a **saturated control** was registered alongside it. The verdict entered is **"criterion not met"** — a statement about the criterion, not a refutation of the hypothesis. What this paper keeps from that batch is the one thing it cannot obtain elsewhere: **a registered cell read under the protocol in force at registration time**, which §7 uses as an instance of "the reported split is not independent" landing on a concrete conclusion.

10. Our own defects (disclosed, not hidden)

#	Defect	Handling
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1	The clean-protocol results document cites a ledger file that does not exist in the sense claimed : neither <code>_r10_final_stats.txt</code> nor the "real log" <code>_r10_stats_0403.txt</code> holds any headline value (6 probes each, 0/6 hits) $\Rightarrow$ no surviving stdout ledger	We do not point at a replacement file (renaming would manufacture a new false citation); the text says "no stdout ledger" , and the p-values are recomputable from <code>pvalue_table_local.py</code> . Detail: §S12
2	the dot15 cell of <code>selection_premium.py</code> was not reproducible $\leadsto$ fixed (stale root directory; the matcher assumed a seed suffix the run names do not carry); it now reproduces M3's +1.12/+0.95/+0.93 value by value	Stated in §5.3; the "3 seeds" note was retracted on 2026-09-21 — the third run's seed is s42, the campaign's fixed shuffle seed. Detail: §S12
3	<code>'sio_b_results.csv'</code> has no written endpoint convention : <code>t2_mask2mende_base100_s42n</code> is recorded as final (67.12) in one	Stated in §7.3, with the consequence of each reading (significant versus not)

	archive and best (70.06) in another	
4	The two headline scripts originally depended on a rented pod (import paramiko plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release	Fixed: both now read the local archive ; the main text uses the local versions and lists the pod versions as cross-checks
5	One script now recomputes the σ decomposition — the data-order and initialization components from the archived per-run values, each asserted against the printed number — but it can only cite the augmentation component, whose record holds an SD and no per-run values; no frozen effect-size rule file ; no script generates 'benchmark_split_audit.md' (hand-written, 28 judgments as literals, though its counts are recomputed)	The first is closed by script, with its one cited (not recomputed) component named; the other two stay labelled "manual / not archived"
6	The local archive has no copy of 6 of the 7 corpora , and of the three-way carve it holds only the manifest (split_defs_20260916/), not the other carve directories	Partially repaired (2026-09-16, GPU run): the 41 runs' per-epoch curves, sio_b_results.csv and the 2×2 evaluation matrix (164 evaluations) are now local; retraining remains non-recomputable
7	The audit document has internal inconsistencies (nine items, found by our own census)	Each adjudicated: 3 misreadings by the inventory, 4 real defects, 1 convention risk, 1 classification correction (§6.4)
8	Our own corpora contain <code>val == test</code>	That is this paper's argument , not an embarrassment to be hidden
9	Our first per-epoch report claimed curve-wide stability from a four-point sample ; the complete curve overturned it for one of the two cells (§8.7)	Corrected in place, with the retracted sentence quoted: the claim is narrowed , not deleted, and the grid's apparent uniformity is named as partly an artefact of where it sampled. Detail: §S12
10	Memory pressure silently changes the training path: under GPU memory pressure Ultralytics switches implementation without changing the recorded configuration — a batch reduction, or a fallback of label assignment to CPU that can fork the trajectory — and case (a) is invisible in the standard artifacts .	Disclosed, not retracted; we do not call the affected batches “clean”. Configuration-level hits: 2 places, 4 runs. We bound it rather than estimate it: ≤ 0.05 pp where the same-config pairs can be checked against each other, and inside the zero-event-control spread elsewhere, so at this scale it is not separable from machine/run variation ; the one clear case is $\Delta V_{\max} + 0.769$ pp. Event counts, the per-run consequences and the log denominators that fingerprint the two modes are in Supplementary S12 (defect 10) .

11. Recommendations

Four checks before reporting, each with the artifact that discharges it. ① Are val and test the same source? Make it a **hard pipeline check** on every dataset YAML (§4.2); the output is the resolved paths. ② If you must select on the reported split, say so and report a selection-free endpoint alongside — the step **enlarges** the arm difference (§5.1). ③ State the test convention: below $n = 6$ a permutation test has no resolution; print the $2/2^n$ floor beside every p. ④ Declare the counting unit behind every count, as Table 3's header does.

⑤ **A plateau is a post-hoc candidate, not a rule we validate.** Weight averaging is established [55];

what is *measured here* averages the top-5 val epochs' **metric values**, capturing **70 %** of the realizable gain against the argmax's **30 %**, at the cost of **0.73 pp** below the peak — a post-hoc reading of our own 40 runs, untested as an intervention. The per-rule table and the candidate mechanism are in **Supplementary S11**, Measurement 3.

12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.
- For several benchmarks the "generic YOLO copy" is **per-project**, not canonical.
- Under the reported unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do not guess.
- Two evidentiary differences (pod versus local) are listed side by side.
- The census is a **literature/release survey**, weaker than our **self-measurement** (§3).
- The decomposition (§5.6) covers **2 of 13 cells**; the others' runs are not on the machine used — a scope we state rather than extrapolate.
- The intervention axis is **one** ($1e0$ **0.001** → **0.005**), and the cross-lineage probe is

one detector, one corpus, six seeds.

- **Four disclosed uncertainties** bound our claims: the 19-row audit is **single-rater**; the log sample is a **convenience sample**; the GPU memory-pressure effect is **bounded, not estimated** (≤ 0.05 pp); and the §8.3 replicates vary the **training file order**, not independent training seeds. No interval folds into §8.6's rate. **Supplementary S12.**

13. Conclusions

This paper's subject is a reporting convention, not a detector, and its premise is that a count is meaningless until its **unit** is named: the same nineteen benchmarks answer differently under each declared unit (Fig. 1), and a second reading of a single row already moves a headline number between 12/19 and 13/19. Three findings stand under that premise. **The most robust number here is the transfer slope κ (§8.7): early `val` movement does transfer to `test`, and the premium is only its residual.** The selection premium is positive in all four cells, and the baseline arm gains more. The paired Δ_{gap} **varies in sign** by corpus, so "the convention understates the effect" is not a claim this paper makes. And on five arm-cells at $n = 10$ the val-selected premium is realized on a disjoint split at **23.3 %** — a run-level descriptive statistic whose cell-level interval includes zero. Two things do **not** hold: the $\sigma\sqrt{2 \ln E}$ scaling was tested and **withdrawn**, and no prospective power analysis was registered, so the nine $n = 3$ cells remain descriptive. What we would ask of a detection paper is narrow: declare the counting unit, freeze in writing which checkpoint is the reported number, and state the interval at the level of the claim. The remediation protocol's first report is the companion's; the measurement under it, and the accounting for where the effect goes, are this paper's.

Declarations

- **Data availability.** Claims are backed by the artifacts in the Supplementary Material; the third-party training logs are **not redistributed** (hashes and source pointers only) and 5 of their sources state no licence. The reproduction package itself — data files, per-epoch matrix, verdict files, checkers, with their SHA-256 list in MANIFEST_sha256.csv — is deposited and openly available at <https://github.com/YBP2005/P2.git> (accessed 2026-09-29).

- **Recomputability boundary.** Every reported number is recomputable from the released per-run artifacts

without retraining, each with its script; only three are not, and they are named: the unredistributable

corpora (§10 defect 6), T1-d's val side ($n = 7$), and the two conflicting-endpoint archives (§7.3).

- **Competing interests.** None declared.

- **One quantity, two spellings.** The $n = 10$ attainable floor is **exactly** $2/2^{10} = 0.001953$; the tables

use the exact value, the abstract writes **0.002**. Same floor, spelling only.

- **CRedit author statement.** *[To be completed with the final author list, using the 14 Elsevier roles.]*

- **Reproducibility.** Every number carries a pointer to a local artifact; **289 audit anchors** mechanically

guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`, `verify_pulled.py`,

`gap_mechanism_20260916.py`, `verify_pr_forms_20260917.py`, `verify_submission_pack_20260917.py`) are

runnable by any holder of the archive. The archive is the one deposited above (**Data availability**).

References

Note on preprint citations. Seventeen of the 55 entries give an arXiv edition. Three of those

also name a peer-reviewed venue ([29] EMNLP-IJCNLP 2019, [46] IEEE TIM 2024, [53] IEEE TNNLS 2026);

four are cited at the canonical release of a benchmark or dataset ([37] COCO, [42] xView, [43] DIOR,

[49] CrowdHuman); two are methodological sources ([26], [27]). The remaining **nine** have no

peer-reviewed version — [6], [10], [52], [54], [17], [15], [55], [21], [18], two from 2025 and seven from

2026. They are the closest concurrent and tooling work, and they are cited **for positioning only**:

no quantitative claim in this paper rests on them, every number here being measured on our own corpora or on the third-party logs of §5.5.

[1] Y.K. Adimoolam, C. Poullis, M. Averkiou. "Data Leakage Detection and De-duplication in Large Scale Geospatial Image Datasets." *CVPR* 2026 (Oral).
https://openaccess.thecvf.com/content/CVPR2026/html/Adimoolam_Data_Leakage_Detection_and_De-duplication_in_Large_Scale_Geospatial_Image_CVPR_2026_paper.html

[2] J. Wang, W. Yang, H. Guo, R. Zhang, G.S. Xia. "Tiny Object Detection in Aerial Images." *2020 25th International Conference on Pattern Recognition (ICPR)* 3791-3798 (2021). DOI 10.1109/icpr48806.2021.9413340

[3] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, et al.. "The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking." *Lecture Notes in Computer Science* 375-391 (2018). DOI 10.1007/978-3-030-01249-6_23

- [4] P.V.A.B. de Venâncio, A.C. Lisboa, A.V. Barbosa. "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices." *Neural Computing and Applications* 34(18):15349-15368 (2022). DOI 10.1007/s00521-022-07467-z
- [5] Z. Zou, K. Chen, Z. Shi, Y. Guo, J. Ye. "Object Detection in 20 Years: A Survey." *Proceedings of the IEEE* 111(3):257-276 (2023). DOI 10.1109/JPROC.2023.3238524
- [6] R. Guedes de Souza, A.R. Panisson. "Who Thinks Best Depends on How Long You Let Them: Budget-Dependent Rankings in LLM Evaluation." arXiv:2608.12150 (2026).
- [7] S. Kapoor, A. Narayanan. "Leakage and the reproducibility crisis in machine-learning-based science." *Patterns* 4(9):100804 (2023). DOI 10.1016/j.patter.2023.100804
- [8] B. Barz, J. Denzler. "Do We Train on Test Data? Purging CIFAR of Near-Duplicates." *Journal of Imaging* 6(6):41 (2020).
- [9] G.C. Cawley, N.L.C. Talbot. "On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation." *Journal of Machine Learning Research* 11(70):2079–2107 (2010).
- [10] A. Apicella, F. Isgrò, A. Pollastro, R. Prevete. "Don't stop me now: Rethinking Validation Criteria for Model Parameter Selection." arXiv:2602.22107 (2026).
- [11] J. Dodge, S. Gururangan, D. Card, R. Schwartz, N.A. Smith. "Show Your Work: Improved Reporting of Experimental Results." *Proceedings of EMNLP-IJCNLP 2019*, pp. 2185–2194. <https://aclanthology.org/D19-1224/>
- [12] K. Gorman, S. Bedrick. "We Need to Talk about Standard Splits." *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*:2786–2791 (2019). DOI 10.18653/v1/P19-1267
- [13] R. Reveles-Martínez, S. Burciaga-Sosa, J.M. Celaya-Padilla, S. Castro-Tapia, H. Luna-García, H. Morales-Magallanes, M.N. Regalado-Pérez, C. Landeros-Soriano, et al.. "Sequence-Aware Dataset Auditing for Leakage-Free Benchmarking of YOLO Detectors for Bottle Detection." *Technologies* 14(9):531 (2026).
- [14] W. Ruangsang, P. Pramkeaw. "Cross-Dataset Evaluation of YOLOv8 for Unmanned Aerial Vehicle Fire and Smoke Detection: Benchmark Contamination, Zero-Shot Transfer, and Onboard Deployment on a Low-Cost Airframe." *Drones* 10(8):635 (2026).
- [15] H. Suo, H. Wang, Y. Li. "Checkpoint Selection and Evaluation in EEG Emotion Recognition." arXiv:2607.27655 (2026).
- [16] Smith JE, Winkler RL. The optimizer's curse: skepticism and postdecision surprise in decision analysis. *Management Science* 52(3):311–322, 2006. doi:10.1287/mnsc.1050.0451.
- [17] S. Roth. "Which Leakage Types Matter? A Quantitative Landscape Across 2,047 Benchmark Datasets." arXiv:2604.04199 (2026).

- [18] M. R. R. Roktim. "When Is a Molecule a Duplicate? Identity Policy Determines What a Benchmark Audit Finds." *ChemRxiv* preprint (2026). DOI 10.26434/chemrxiv.15009099/v1
- [19] G. Jocher, A. Chaurasia, J. Qiu. *Ultralytics YOLO* (v8.4.120) [computer software] (2023; pinned release 2026). <https://github.com/ultralytics/ultralytics> (accessed 2026-09-12); the same project's official course states the split-independence rule normatively: Ultralytics Academy, "Splits that Tell the Truth," <https://academy.ultralytics.com/courses/computer-vision-foundations/splits-that-tell-the-truth> (accessed 2026-09-26). [verified: software release tagged v8.4.120 as used in this study]
- [20] B. Recht, R. Roelofs, L. Schmidt, V. Shankar. "Do ImageNet Classifiers Generalize to ImageNet?" *Proceedings of the 36th International Conference on Machine Learning*, PMLR **97**:5389–5400 (2019).
- [21] G. Zhang, K. Zhao. "Winning by Peeking: Unenforced Budgets and Test-Set Selection Inflate Short-Budget AutoML Comparisons." arXiv:2608.07303 (2026).
- [22] Taylor J, Tibshirani RJ. Statistical learning and selective inference. *Proceedings of the National Academy of Sciences* **112**(25):7629–7634, 2015. doi:10.1073/pnas.1507583112.
- [23] Benjamini Y, Yekutieli D. False discovery rate—adjusted multiple confidence intervals for selected parameters. *Journal of the American Statistical Association* **100**(469):71–81, 2005. doi:10.1198/016214504000001907.
- [24] M. Everingham, L. Van Gool, C.K.I. Williams, J. Winn, A. Zisserman. "The Pascal Visual Object Classes (VOC) Challenge." *International Journal of Computer Vision* 88(2):303-338 (2009).
- [25] Mendeley Data records, "Face Mask Detection" (data records, no venue). Widely used 853-image PASCAL-VOC variant: <https://www.kaggle.com/datasets/andrewmvd/face-mask-detection>
- [26] J. Dodge, G. Ilharco, R. Schwartz, A. Farhadi, H. Hajishirzi, N. Smith. "Fine-Tuning Pretrained Language Models: Weight Initializations, Data Orders, and Early Stopping." arXiv:2002.06305 (2020). DOI 10.48550/arXiv.2002.06305
- [27] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Sepah, E. Raff, et al. "Accounting for Variance in Machine Learning Benchmarks." arXiv:2103.03098 (2021). DOI 10.48550/arXiv.2103.03098
- [28] B.L. Welch. "The Generalization of 'Student's' Problem when Several Different Population Variances are Involved." *Biometrika* 34(1/2):28 (1947).
- [29] K. Kann, K. Cho, S.R. Bowman. "Towards Realistic Practices In Low-Resource Natural Language Processing: The Development Set." *Proceedings of EMNLP-IJCNLP 2019*; arXiv:1909.01522 (2019).

- [30] R. Roelofs, V. Shankar, B. Recht, S. Fridovich-Keil, M. Hardt, J. Miller, L. Schmidt. "A Meta-Analysis of Overfitting in Machine Learning." *Advances in Neural Information Processing Systems* **32**:9179-9189 (2019).
- [31] Y. Benjamini, Y. Hochberg. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *Journal of the Royal Statistical Society Series B: Statistical Methodology* **57**(1):289-300 (1995).
- [32] *YOLOX* [computer software]. <https://github.com/Megvii-BaseDetection/YOLOX> (pinned at commit 6ddff4824372, branch main, accessed 2026-09-16).
- [33] *Detectron2* [computer software]. <https://github.com/facebookresearch/detectron2> (pinned at commit a2f4a8771ab7, branch main, accessed 2026-09-16).
- [34] *MMDetection* [computer software]. <https://github.com/open-mmlab/mmdetection> (pinned at commit cfd5d3a985b0, branch main, accessed 2026-09-16).
- [35] *DETR* [computer software]. <https://github.com/facebookresearch/detr> (pinned at commit 29901c51d7fe, branch main, accessed 2026-09-16).
- [36] *PaddleDetection* [computer software]. <https://github.com/PaddlePaddle/PaddleDetection> (pinned at commit b25522a0f4bd, branch release/2.9, accessed 2026-09-16).
- [37] Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, Piotr Dollár. "Microsoft COCO: Common Objects in Context." arXiv:1405.0312 (2014). DOI 10.48550/arXiv.1405.0312
- [38] S. Shao, Z. Li, T. Zhang, C. Peng, G. Yu, X. Zhang, J. Li, J. Sun. "Objects365: A Large-Scale, High-Quality Dataset for Object Detection." *2019 IEEE/CVF International Conference on Computer Vision (ICCV)* 8429-8438 (2019).
- [39] A. Kuznetsova, H. Rom, N. Alldrin, J. Uijlings, I. Krasin, J. Pont-Tuset, S. Kamali, S. Popov, et al.. "The Open Images Dataset V4." *International Journal of Computer Vision* **128**(7):1956-1981 (2020).
- [40] G.-S. Xia, X. Bai, J. Ding, et al. "DOTA: a large-scale dataset for object detection in aerial images." *CVPR* 2018, pp. 3974-3983. DOI 10.1109/CVPR.2018.00418.
- [41] P. Zhu, L. Wen, D. Du, et al. "Detection and tracking meet drones challenge." *IEEE TPAMI* **44**(11):7380-7399 (2022). DOI 10.1109/TPAMI.2021.3119563. (VisDrone.)
- [42] Darius Lam, Richard Kuzma, Kevin McGee, Samuel Dooley, Michael Laielli, Matthew Klaric, Yaroslav Bulatov, Brendan McCord. "xView: Objects in Context in Overhead Imagery." arXiv:1802.07856 (2018). DOI 10.48550/arXiv.1802.07856
- [43] Ke Li, Gang Wan, Gong Cheng, Liqiu Meng, Junwei Han. "Object Detection in Optical Remote Sensing Images: A Survey and A New Benchmark." arXiv:1909.00133 (2019). DOI 10.48550/arXiv.1909.00133 (DIOR benchmark)

- [44] G. Cheng, J. Han, P. Zhou, L. Guo. "Multi-class geospatial object detection and geographic image classification based on collection of part detectors." *ISPRS Journal of Photogrammetry and Remote Sensing* 98:119-132 (2014).
- [45] Xu et al. "Detection of safety helmet wearing based on improved SSD." *Journal of Physics: Conference Series* (2019). Dataset: <https://github.com/njvisionpower/Safety-Helmet-Wearing-Dataset>
- [46] Fusheng Yu, Jiang Li, Xiaoping Wang, Shaojin Wu, Junjie Zhang, Zhigang Zeng. "Large, Complex, and Realistic Safety Clothing and Helmet Detection: Dataset and Method." arXiv:2306.02098 (2023); journal version in *IEEE Transactions on Instrumentation and Measurement* (2024). DOI 10.48550/arXiv.2306.02098 (SFCHD / SFCHD-SCALE)
- [47] S. Ge, J. Li, et al. "Detecting masked faces in the wild with LLE-CNNs." *CVPR* 2017, pp. 426-434. DOI 10.1109/CVPR.2017.53. (Cited only to delimit the MAFA packaging difference.)
- [48] S. Yang, P. Luo, C.C. Loy, X. Tang. "WIDER FACE: A Face Detection Benchmark." *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 5525-5533 (2016).
- [49] Shuai Shao, Zijian Zhao, Boxun Li, Tete Xiao, Gang Yu, Xiangyu Zhang, Jian Sun. "CrowdHuman: A Benchmark for Detecting Human in a Crowd." arXiv:1805.00123 (2018). DOI 10.48550/arXiv.1805.00123
- [50] L. Cascone, M. Nappi, C. Pero, X. Wang. "A framework for bias-aware dataset evaluation in soft facial attribute recognition." *Pattern Recognition* 172:112416 (2026). DOI 10.1016/j.patcog.2025.112416
- [51] Y. Huang, Z. Zhang, Y. Huang, Q. Wu, H. Huang, Y. Zhong, L. Wang. "Customized meta-dataset for automatic classifier accuracy evaluation." *Pattern Recognition* 146:110026 (2024). DOI 10.1016/j.patcog.2023.110026
- [52] J. Sweeney. "Optimizer Memory Makes Shuffle Order a First-Order Source of Fine-Tuning Noise." arXiv:2606.29554 (2026).
- [53] E. Pintelas, I.E. Livieris. "GeNeX: Genetic Network eXperts framework for addressing Validation Overfitting." *IEEE Transactions on Neural Networks and Learning Systems* (2026); arXiv:2603.11056. DOI 10.48550/arXiv.2603.11056
- [54] Y. Tian, Q. Ye, D. Doermann. "YOLOv12: attention-centric real-time object detectors." arXiv:2502.12524 (2025).
- [55] N. Ajroldi, A. Orvieto, J. Geiping. "When, Where and Why to Average Weights?" arXiv:2502.06761 (2025).